

P6.2 Uses and hazards

Summary

We now address the hazards and applications of radioactive decay. The processes of fission and fusion as a source of energy are also considered.

Underlying knowledge and understanding

Learners may have prior understanding of the term radioactivity from the previous sub topic and may be familiar with some uses, but will not have covered this content prior to this topic.

Common misconceptions

Learners tend to think that radioactivity will always cause physical mutations when humans or animals come into contact with it. They tend to only think of the negative impacts of radiation and not the positive uses.

Tiering

Statements shown in **bold** type will only be tested in the Higher Tier papers. All other statements will be assessed in both Foundation and Higher Tier papers.

Topic content		Opportunities to cover:		Practical suggestions
Learning outcomes	To include	Maths	Working scientifically	
P6.2a recall the differences between contamination and irradiation effects and compare the hazards associated with these two			WS1.1a, WS1.1b, WS1.2a, WS1.2d, WS1.3b, WS1.3f	Use of spark chamber to demonstrate a different type of activity counter.
P6.2b ☒ explain why the hazards associated with radioactive material differ according to the half-life involved			WS1.1a, WS1.1c, WS1.1d, WS1.1e, WS1.1f, WS1.1h, WS1.1i	Illustrate an everyday use of radioactive sources in smoke detectors and discuss why they might be suitable.

P6.2c ☒	describe the different uses of nuclear radiations for exploration of internal organs, and for control or destruction of unwanted tissue			WS1.1a, WS1.1c, WS1.1d, WS1.1e, WS1.1f, WS1.1h, WS1.1i	Research the medical uses of radioactive tracers and radiotherapy.
P6.2d ☒	recall that some nuclei are unstable and may split, and relate such effects to radiation which might emerge, to transfer of energy to other particles and to the possibility of chain reactions	knowledge of the term nuclear fission for fission to occur the unstable nucleus must usually first absorb a neutron			
P6.2e ☒	describe the process of nuclear fusion	knowledge that mass may be converted into the energy of radiation			